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Social Construction of Invention through Patent Management: Thomas Blanchard's Woodworking Machinery

CAROLYN C. COOPER

Invention in the United States in the 19th century was to a significant degree socially constructed through patent management. By social construction of invention I mean not only the social shaping of *specific* inventions but also, at a deeper level, the determining of the very rules by which people defined “newness” in inventions. Since originality is the defining characteristic of any invention, the gradual social formation of decision rules for originality was tantamount to defining invention itself.

Of course, we recognize that not all inventions received patents. Still, all patents were, by definition, for inventions. Publicly acknowledged patented inventions provided powerful standards for recognition of nonpatented inventions also. In the course of managing patents, persons interacted so as to constitute an important social subsystem to which our society allocated the continuing task of defining “invention,” that is, originality or newness in man-made objects. These interactions determined features of particular patented devices, *and* over time they defined and redefined implicit rules for judging whether a proposed invention was really new.

The Rationale and Evolution of a Patent System

Thomas Blanchard (1788–1864) was a Massachusetts farm boy who became an inventor of machinery, most notably the irregular turning lathe. He was able to make a career of inventing because the society in which he lived—the 19th-century United States—gave him

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material reward as well as fame for his acts of invention.¹ Not all societies have so rewarded their inventors, but modern Western ones have tended to do so.² From this we can infer that these societies have generally approved of invention and intended to encourage it. From the point of view of a society intending to encourage invention by rewarding inventors, it is important that the reward be sufficient for such encouragement but not excessive. It is also important that particular inventions, once made, should not disappear from view but should become part of the society's available stock of technology. Thus, the optimal social system to satisfy these criteria is one that holds out the maximum credible prospect of reward to a potential inventor to induce him to make an invention and to share it with society, but one that then gives him the minimum actual reward consistent with continued credibility.

Sometimes, as in prerevolutionary France, modern Western societies have chosen to award prizes out of public funds directly to the inventors who have persuaded designated judges (such as the Académie des Sciences in Paris) that their invention deserves approval.³ Since the early 17th century, however, the usual English and American way of rewarding invention has been less direct: these societies have given inventors an opportunity to earn reward through the private economy, by granting them a patent—a temporary monopoly on the making, use, and sale of a new device—in return for describing it so fully that anyone “skilled in the art” can understand it.⁴ The patent has thus acquired two functions: one is to define and publicize what is original about an invention, and the other is to turn the

¹In addition to money and fame, Fritz Machlup has identified three other motivations for invention: fun, service to mankind, and Veblen's “instinct of workmanship.” See “The Supply of Inventors and Inventions,” in *The Rate and Direction of Inventive Activity: Economic and Social Factors*, ed. Richard R. Nelson (Princeton, N.J., 1962), p. 144. In the present discussion, money and fame are what the social system can provide directly as rewards for invention; the other motivations are possibly also social in origin but are not at issue here.

²For the slow emergence of reward for invention out of royal grants of privilege, see Maximilian Frumkin, “Early History of Patents for Invention,” *Transactions of the Newcomen Society* 26 (1947–49): 47–56.

³Roger Hahn, *The Anatomy of a Scientific Institution: The Paris Academy of Sciences, 1666–1803* (Berkeley and Los Angeles, 1971), pp. 66–70.

⁴The beginning of patents for invention in England is usually dated to the Statute of Monopolies in 1624. Two excellent studies of the English patent-management system are Christine MacLeod, *Inventing the Industrial Revolution: The English Patent System, 1660–1800* (Cambridge, 1988); and Harold I. Dutton, *The Patent System and Inventive Activity during the Industrial Revolution, 1750–1852* (Manchester, 1984). For a brief account of “the origin and growth of the patent system in Britain,” see Allan Gomme, *Patents of Invention* (London, 1952).

invention for a limited duration of time into a piece of private property over which the owner has legal rights.

An inventor in possession of a United States patent in the 19th century found it required management if it was to yield a return. He (or—rarely—she) had entered the patent-management system, in which patentees could *use* their own patents to produce something, or they could *sell* or *license* them to other people. They also had to *defend* their patents in lawsuits against infringers. Before 1861 they might also restate the specification and get it *reissued* for the same patent term or attempt to *extend* their patent term. These were all patent-management activities. Some inventors, like Thomas Blanchard, or Thomas Edison in a later generation, were relatively successful in managing their patents; some inventors, like Azariah Woolworth (and presumably many others of whom we have never heard), were poor at patent management and therefore dropped out of history. By keeping records that necessarily emphasize success in patent management rather than some unattainably pure, objective account of invention, the patent system has encouraged historians to create an oversimplified history of technological change as large discontinuous leaps forward instead of continual, small, evolutionary modifications.

To help inventors obtain and manage patents, the occupational specialties of patent agent and patent attorney emerged. For a fee, they would advise inventors as to the patentability of their inventions and represent them in dealing with the Patent Office and in lawsuits. Patent agents and agencies such as Munn & Co., associated with the publication *Scientific American*, appeared in New York, Washington, D.C., and other large and small towns.⁵ As figure 1 shows, at least one midwestern agent late in the century also went on the road to drum up business in rural areas.

The United State patent system itself underwent changes in its rules and procedures.⁶ From its establishment in 1790, it required inventors to reveal information about their inventions through written patent

⁵By 1870 Munn & Co. was boasting that they had handled nearly 20,000 patents in over twenty years of experience: Munn & Co., *The United States Patent Law* (New York, 1871), p. 17. Another agency in Washington, D.C., in the mid-19th century was associated with John J. Greenough's shorter-lived *Polytechnic Journal*. See Robert C. Post, *Physics, Patents, and Politics: A Biography of Charles Grafton Page* (New York, 1976), p. 160.

⁶For useful brief accounts of these changes, see Morgan Sherwood, "The Origins and Development of the American Patent System," *American Scientist* 71 (September–October 1983): 501–6; H. A. Meier, "Thomas Jefferson and a Democratic Technology," in *Technology in America*, ed. C. W. Pursell (Cambridge, Mass., 1981), p. 29; and Nathan Reingold, "U.S. Patent Office Records as Sources for the History of Invention and Technological Property," *Technology and Culture* 1 (1960): 156–67. For more detail,

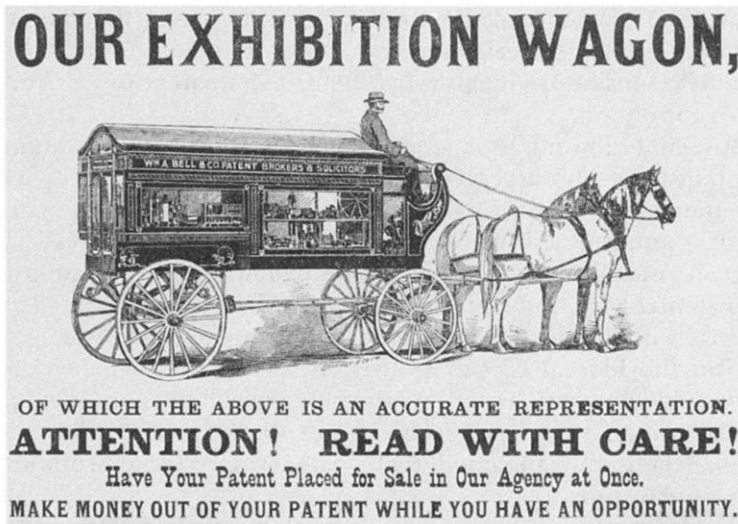


FIG. 1.—In 1891 William A. Bell & Co., patent brokers and solicitors of Sigourney, Iowa, outfitted a wagon for mobilized display of up to 150 patent models and copies of patent specifications. It toured midwestern states in the summer and fall and southern states the following winter. Bell & Co. told inventors that besides taking in state fairs and other large expositions, the “agents having the wagon in charge . . . call on Manufacturers, Speculators and Buyers. . . . Do not miss this good opportunity of having your patent brought before those most likely to buy territory.” (Warshaw Collection, National Museum of American History.)

specifications and drawings and, since 1836, by submitting models as part of the application. (Hence the models in the exhibit in fig. 1.) The requirement of models ended in 1880.⁷ The occupation of patent examiner evolved within the patent system. For a brief period after 1790 Secretary of State Thomas Jefferson served as the first American patent examiner; then in 1793 the United States adopted a simple registration system for patents, such as England had. It was

see P. J. Federico, ed., “Outline of the History of the United States Patent Office,” *Journal of the Patent Office Society* 17, no. 7 (Centennial Number) (July 1936).

⁷See Kendall J. Dood, “Patent Models and the Patent Law: 1790–1880,” *Journal of the Patent Office Society* 65, nos. 4 and 5 (April and May 1983): 187–216, 234–74, “Why Models?” in *American Enterprise: Nineteenth-Century Patent Models*, ed. Robert C. Post (New York, 1984), and “Patenting and Patent Models in Nineteenth-Century America,” in *Icons of Invention: American Patent Models*, ed. Barbara Suit Janssen (Washington, D.C., 1990), pp. 11–13. For the vicissitudes of making and housing models, see William Ray and Marlys Ray, *The Art of Invention: Patent Models and Their Makers* (Princeton, N.J., 1974).

left to each inventor to swear that his invention was original and left to the legal system to settle conflicts of opinion on that point. In 1794, for instance, Jefferson had been relieved of the duty to examine claims of originality when he issued a patent to Eli Whitney for his cotton gin.

Between 1793 and 1836, then, there were no patent examiners, only registrars who accepted the applications and \$30 fees—which were much lower than English patent fees—and issued the patents after a waiting period in which any other claimant could object and bring an “interference suit” within the patent system. A patent gave the patentee the right to sue anyone who used his intellectual property without paying him. These “infringement suits” usually took place in the federal Circuit Courts. Before 1836 the inventors of nearly 10,000 inventions, among them Thomas Blanchard, paid their \$30, submitted their specifications and drawings and (sometimes) models, then duly obtained their patents and began conducting infringement suits.

After forty-odd years experience with this register-and-sue system, which became increasingly unwieldy as the number of patents swelled, Congress took action in 1836 to relieve the courts and put the Patent Office on a more professional footing. In the reorganized Patent Office, salaried patent examiners would advise the patent commissioner to issue a patent only if they judged the application to indicate a truly original invention.

Senator John Ruggles of Maine, promoter of the Patent Office reform and recipient of the first patent under the new system, said an examiner should have “a general knowledge of the arts, manufactures, and the mechanisms used in every branch of business in which improvements are sought to be patented, and of the principles embraced in the ten thousand inventions patented in the United States, and of the thirty thousand patented in Europe. He must moreover possess a familiar knowledge of the statute and common law on the subject, and the judicial decisions both in England and our own country, in patent cases.”⁸ Senator Ruggles acknowledged that men meeting all these criteria were rare. Nevertheless, from one patent examiner in 1836 and two in 1837, the number grew by 1861 to twelve examiner positions and by 1870 to twenty-two principal examiners plus forty-four assistant examiners.⁹ (See fig. 2.)

The patent examiners’ decisions were ostensibly objective, but inevitably the examiners differed among themselves in the degree of

⁸Post, *Physics, Patents, and Politics*, p. 51.

⁹Ibid.; also Munn & Co. (n. 5 above), p. 52.

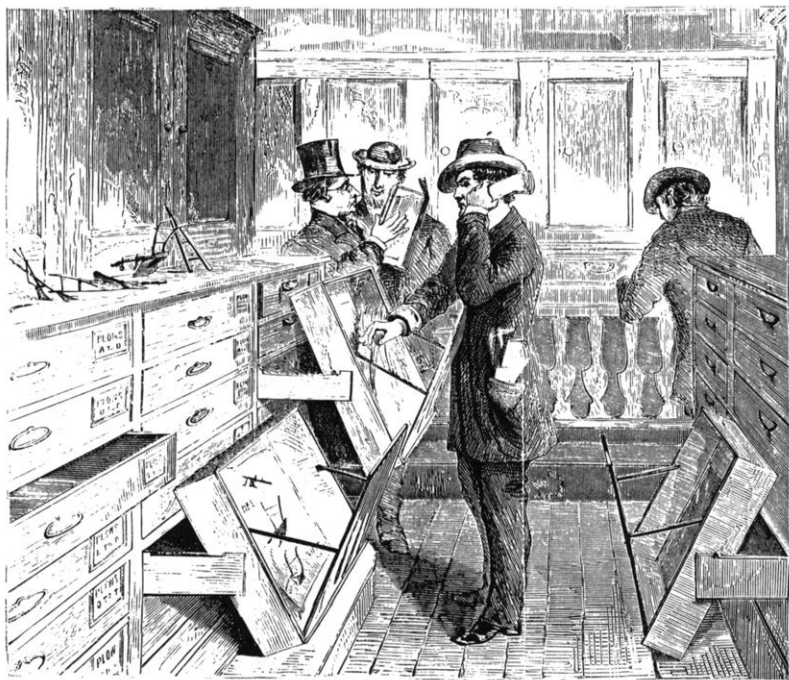


FIG. 2.—Patent examiners consulted models and drawings (here, for plows) as well as written specifications in deciding whether inventions were truly original. (*Harper's Weekly Magazine*, July 10, 1869, p. 445.)

originality they thought would deserve a patent.¹⁰ No one had spelled out their decision rules in any formal way; they were left to construct them in practice. Robert C. Post's study of the rejection rate for antebellum patent applications demonstrates that, as a group, the examiners changed over time in the stringency with which they applied their implicit decision rules.¹¹ We may suppose they formed these implicit rules in communication with one another as well as with inventors, patent agents, and other people in the growing community of persons interested in mechanical inventions. Sometimes patent examiners became patent agents themselves, and vice versa.

To encourage the decorative arts as well as utilitarian inventions, in 1842 Congress passed a law to expand the category of patentable entities to include not only products and processes, but also decorative designs. The same considerations of examination, property rights,

¹⁰Post, *Physics, Patents, and Politics*, pp. 50–51.

¹¹Robert C. Post, " 'Liberalizers' versus 'Scientific Men' in the Antebellum Patent Office," *Technology and Culture* 17 (1976): 24–54.

and infringements applied for design patents as for “utility” patents, except that initially the term was only seven instead of fourteen years.¹² In both instances, the examiners communicated with the inventors or with their patent agents about *which* features of a proposed invention were new and which already existed. Then the inventors and patent agents could rewrite (or redraw) the specification to disclaim the originality of some features and claim others. They could resubmit the application for decision again and again. Eventually, either the patent was granted or the applicant gave up.¹³

As Kendall Dood points out, even after a patent was granted, if a patentee decided—perhaps in the course of a lawsuit—that the wording of his specification had been inaccurate or unclear, he could withdraw his patent, reword the specification, and apply to get the patent *reissued* for the remainder of the original patent term. The Patent Act of 1836 also allowed for a seven-year *extension* of the term of a patent if the patentee failed to reap “a reasonable remuneration for the time, ingenuity and expense bestowed upon [his invention], and the introduction thereof into use.”¹⁴ The patented invention was supposed to remain the same in an extension or a reissue; in fact, however, these provided unintended occasions for making changes in the scope of a patent after experience with it. Reissues and extensions were much diminished by changes in Patent Office rules in 1861 and 1870.¹⁵

Patent Litigation

Additional occasions for changes in a patented invention were provided by lawsuits, for the Patent Act of 1836 did not eliminate recourse to the legal system. It just meant that now the originality of inventions was scrutinized and socially determined at two locations in society—the Patent Office and the courts. In individual cases the judgment of the courts could, implicitly, overrule the judgment of the patent examiners in having granted a patent. Conducting lawsuits remained a task of patent management, for the more useful an invention, the more likely there was to be infringement and therefore

¹²Patent Act of August 29, 1842 (5 Stat. at Large 543). By 1870, design patents were available for 3½, 7, or 14 years, for respective fees of \$10, \$15, or \$30.

¹³Reingold (n. 6 above), p. 160, likens the patent application process to an editorial conference between a writer and editor before publication.

¹⁴Patent Act of July 4, 1836 (5 Stat. 117, sec. 18), quoted in Reingold, p. 163.

¹⁵On reissues, see Kendall J. Dood, “Pursuing the Essence of Inventions: Reissuing Patents in the 19th Century,” in this issue. By 1870 the term of a patent was lengthened from fourteen to seventeen years and was no longer extendable by Patent Office procedures, although an act of Congress could still do so.

litigation. Although deplored, lawsuits were a normal feature of the system, not a symptom of its malfunctioning nor of cranky personal temperaments.

At stake in the lawsuits was not just the original inventor's claim to fame, but the economic wherewithal of a network of persons—the assignees and licensees to whom he may have sold or licensed his intellectual property for a given purpose in different geographic territories or for different purposes in the same territory. If no one sued infringers, who had avoided paying royalties for using the patented device, the assignees and licensees would be at a disadvantage in competing with them. Without infringement suits, an inventor's intellectual property would be eroded and gradually rendered worthless.

Infringement suits were usually held in federal Circuit Courts, presided over by circuit-riding Supreme Court justices. Sometimes the judges decided the case, sometimes a jury. One defense against a complaint of infringement was that the defendant's device was substantially different from the patented one. Another defense was that the plaintiff's patented device was not in fact original, but the same as a previous device. Or both of these arguments might be used at once. For either argument, it was necessary to make comparisons of two objects at a time: the plaintiff's device was compared with some device in the past, or it was compared with the defendant's device. More-or-less expert witnesses on both sides would identify various features of the devices as materially different from each other or substantially the same. As with the Patent Office examiners, there were no objective and explicit decision rules for these witnesses to follow in making comparisons of one device with another; they simply had to use their own perceptions. After hearing their testimony, it was up to the judges or jury to decide whether an infringement had occurred.

Thus, in the granting and maintaining of patents, the nub of the process was to get people—examiners in the Patent Office or witnesses in lawsuits—to compare two things and say yes, gizmo x and y are the same in relevant respects, or no, gizmo x is significantly different from gizmo y.¹⁶ In both contexts we may observe these comparisons

¹⁶Citing Wittgenstein, Trevor Pinch recently expressed this point: "a relationship of similarity or . . . difference does not rest out there waiting for us to find it, but depends upon [our] classifying two things as similar or different. In order to say two things are similar . . . we select from the myriad of possibilities the relevant properties whereby we judge two things to be similar"; Trevor Pinch, "Testing—One, Two, Three . . . Testing!" *Testing: A Test Case for the New Sociology of Technology*, working paper FS II 90–502 (Wissenschaftszentrum Berlin für Sozialforschung, Berlin), pp. 9–10.

that were made in the past and try to discern what the criteria were for the decisions.¹⁷ In making decisions the examiners and witnesses were of course interacting directly or indirectly with judges, juries, lawyers, patent agents, and inventors, for the decisions of the courts were known and responded to by the patent examiners; the decisions of patent examiners had weight in the decisions of the courts. The lawyers marshaling evidence on both sides of these cases encouraged witnesses to make comparisons on grounds that had been convincing to judges and juries in previous trials, so there tended to be a rough convergence over time as to what were the relevant grounds for comparing one device to another. Patent agents and attorneys advised the inventors, and so on. All of these people were actors in the social system that was continuously creating and shaping an inventor's intellectual property, both before and after the initial granting of the patent.

So, through the interplay of the two functions of patents—as disclosure, or description, of the invention and as private property—inventions in 19th-century America were socially defined and redefined. The material devices themselves changed as their features were modified during different uses, and unintended opportunities arose for official redefinition of the invention, when patents were reissued or extended in term or litigated. Then, since the Patent Office and the courts kept records of these actions, subsequent inventors who wanted patents were able to get information about previous inventions and to build on them, being of course careful not to imitate them too closely. How close was “too close” became a matter of social construction through patent management.

Patented Flatware Designs

To see how this process worked, let us first consider an example with which we are familiar in everyday life—the decorative patterns on the handles of knives, forks, and spoons. The shift in production technology from individual handcrafting to quantity factory production of metal flatware took place in the 19th century with the development of diesinking techniques and stamping machines capable of striking out flatware blanks with identical raised and sunk patterns. Quantity production was also much fostered by the development in midcentury of electroplating of silver onto patterned

¹⁷These comparisons are observable in the “patented” patent application files and in patent litigation records, both held in various locations by the National Archives. For procedures in finding litigation records, see Carolyn C. Cooper, “Thomas Blanchard’s Woodworking Machines: Tracking 19th-Century Technological Diffusion,” *IA: The Journal of the Society for Industrial Archeology* 13, no. 1 (1987): 41–54.

cutlery made of base metals.¹⁸ Hence the former silversmithing tasks were divided among diesinkers, designers, and manufacturers. After 1842 designers could patent their decorative patterns and sell them to manufacturers, who would then get the diesinkers to copy the new patterns into dies. Making the steel dies was expensive; once a manufacturer had begun production of an attractive design, he did not like to find that another manufacturer had copied it and was stealing his customers. With possession of a patented design, he had recourse to the courts in such cases. In the twenty years from 1842 to 1861 only thirty-four design patents for flatware were issued, but then the rate rose sharply to 100 issued in the eleven years, 1862–73.¹⁹

Three men otherwise known as the Gorham Silver Co. of Providence, Rhode Island, received a flatware design patent in 1861.²⁰ Several years later George C. White began making and selling flatware with designs patented to Leroy S. White of Waterbury, Connecticut.²¹ In 1870 the Gorham Co. sued White for patent infringement. Figure 3 shows Gorham's and White's patented designs. The drawing for a design patent, unlike that for a "utility" patent, tells you all there is to know about it. Are White's designs the same as Gorham's or different?

Testimony on both sides in this case came from silverware dealers and manufacturers, diesinkers, engravers, designers, jewelers, patent agents, and "editors of scientific publications"—a spectrum of persons "skilled in the art" of making flatware and patent decisions. For instance, C. L. Tiffany, founder and head of the house of Tiffany and Co. in New York, who had dealt in forks and spoons for more than twenty-five years, said he had "no hesitation in saying they are substantially alike. I think the resemblance would mislead ordinary purchasers; and being asked I certainly might myself be misled by it,

¹⁸Stephen K. Victor, "'From the Shop to the Manufactory': Silver and Industry, 1800–1970," in *Silver in American Life*, ed. Barbara M. Ward and Gerald W. R. Ward (Boston, 1979), pp. 23–32; Dorothy T. Rainwater and H. Ivan Rainwater, "Patented Processes," chap. 4 in their *American Silverplate* (Nashville, 1968); Deborah D. Waters, "From Pure Coin: The Manufacture of American Silver Flatware," *Winterthur Portfolio* 12 (1977): 19–33.

¹⁹These are listed in M. D. Leggett, comp., *Subject Matter Index of Patents for Inventions, 1790–1873* (Washington, D.C., 1874), vol. 3, in "Index of Designs Issued from the United States Patent Office from 1790 [*sic*] to 1873, Inclusive," under "cutlery," "fork," "knife," "spoon," and their "handles," pp. 1872, 1874–75, 1878, 1899–91, 1905–6.

²⁰John Gorham, Gorham Thurber, and Lewis Dexter, Jr., U.S. Design Patent no. 1,440, July 16, 1861.

²¹Leroy S. White, U.S. Design Patent no. 2,551, January 15, 1867, and no. 2,992, March 31, 1868.

GORHAM CO.

WHITE, 1867.

WHITE, 1868.

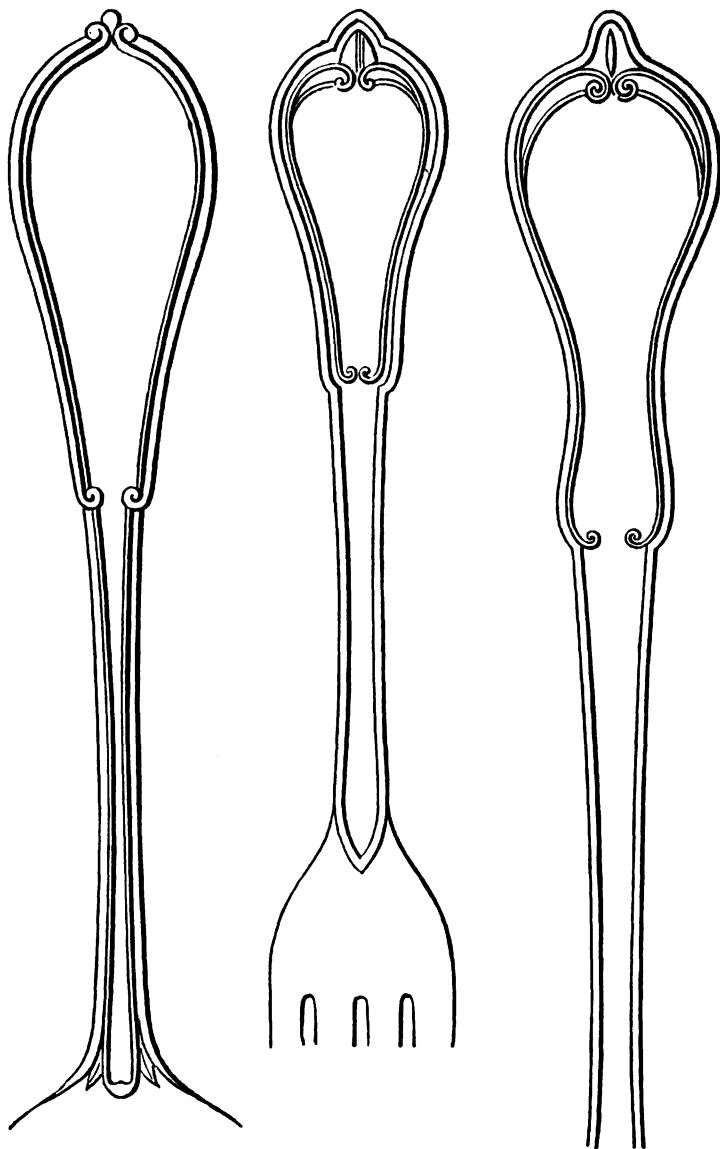


FIG. 3.—Gorham Silver Co., whose flatware design at left was patented in 1861, sued George C. White for infringement when he began manufacturing flatware with the designs shown in middle and at right, patented in 1867 and 1868. (81 U.S. Reports: 521.)

if not beforehand told of the difference and my attention particularly called to it.”²²

However, Henry Ball, identified as the senior member of the well-known firm of Ball, Black & Co., New York City, a silversmith and jeweler who had been in this business since 1832, testified that the patterns were “substantially different, both in shape and design.”²³ The Circuit Court for the southern New York district heard these and other witnesses and ruled that the patterns were different enough—at least in the eyes of witnesses who were knowledgeable about silverware—that White’s design was not an infringement of Gorham’s.²⁴

But the case was appealed to the Supreme Court, which ruled that White’s design was likely to be confused with Gorham’s by ordinary, nonexpert persons who were, after all, the customers. So it reversed the lower court’s decision, but only over dissenting votes by three of the justices.²⁵ This case was much cited as precedent in design-patent suits later, even for carpets or other nonmetallic objects. When called on to make comparisons, however, more-or-less expert witnesses and judges continued to dispute and to dissent, for their own eyes saw clear differences or deceptive resemblances in designs. They made decisions—how else could they?—by vote.

Such lawsuits were occasions for social delineation of the boundaries on intellectual property. Compared with patents for manufacturing devices, the patented designs are cases in which we can appreciate more easily that there were no objective characteristics that made them unambiguously “different from” or “the same as” one another, except as perceived by the eye of their beholder. But the same social process took place in infringement suits for utility patents as well, as the example of Thomas Blanchard’s irregular turning lathe demonstrates.

The Blanchard Lathe

The gunstocking lathe for which Blanchard obtained a patent in 1819 copied a model all around, as distinct from a modern-day key-copying machine, which copies a key only along its edge. It was different from an ordinary lathe, which could only shape objects that were circular in cross-section. It was also different from special ornamental turning lathes, which could copy so-called rosette patterns and irregular bas-reliefs, like medallions or coins, onto the surface of disks or cylinders. Both ordinary and ornamental lathes cut

²²81 U.S. Reports: 514.

²³Ibid., p. 518.

²⁴10 Federal Cases: 827–30.

²⁵81 U.S. Reports: 524–31.

by holding a cutting point to a rapidly rotating workpiece, so the cutting motion is the motion of the object itself. That is why they can only cut circular cross-sections around the axis of rotation. But in Blanchard's machine a *rapidly* rotating cutting wheel, guided by another wheel that traced the contours of a *slowly* rotating model, removed material from all around a slowly rotating workpiece, down to the shape of a model, even one whose shape was that of a gunstock or irregular object *not* circular in cross-section. (See fig. 4.) It took just a few minutes to accomplish this in one continuous operation.

As an "inside contractor," Blanchard used his lathe in the 1820s at the national armory in Springfield, Massachusetts, to make gunstocks for military muskets. It is still preserved in the museum there, and a full-size replica is on display at the National Museum of American History in Washington, D.C. It was a versatile machine, for with relatively little effort one could set it up to make a different object by use of a different model. Besides gunstocks, Blanchard and his assignees and licensees adapted the machine to make shoe lasts, hat blocks, ax helves and other tool handles, and wheel spokes and other carriage parts, all of which are irregular and previously had to be made by hand tools. The patent drawing (fig. 5) shows the lathe adapted to turning shoe lasts. A last maker could make three to six pairs of shoe lasts a day by hand; that is, he took almost an hour to two hours per last.²⁶ But even the earliest Blanchard lathe could rough-shape uniform shoe lasts or gunstocks in four minutes each. Smooth turning took only nine minutes. With its uniformity of output and much greater speed, Blanchard's irregular turning lathe provided a technical precondition for industrializing the large-scale manufacture of shoes, hats, agricultural equipment, and road vehicles, as well as small arms. It was a very important invention.

But recognition did not come automatically, nor did reward. They came only with patent management. Although he invented many other machines as well, Blanchard's experience with this machine for the rest of his life provides an excellent example of all the actions an inventor of his day could take in relation to the patent-management system. In attempting to gain reward for his invention, Blanchard not only used his own patent to make and sell machines, but also managed to overcome problems posed by interference, reissue, infringement, extension, assignment, and license. In addition, his management activities involved congressional lobbying in Washington. Blanchard's experience as patent manager was not that of the average inventor,

²⁶A range of 600–720 minutes in a ten- or twelve-hour day divided by 6 to 12 lasts equals a range of 50 to 120 minutes per last.

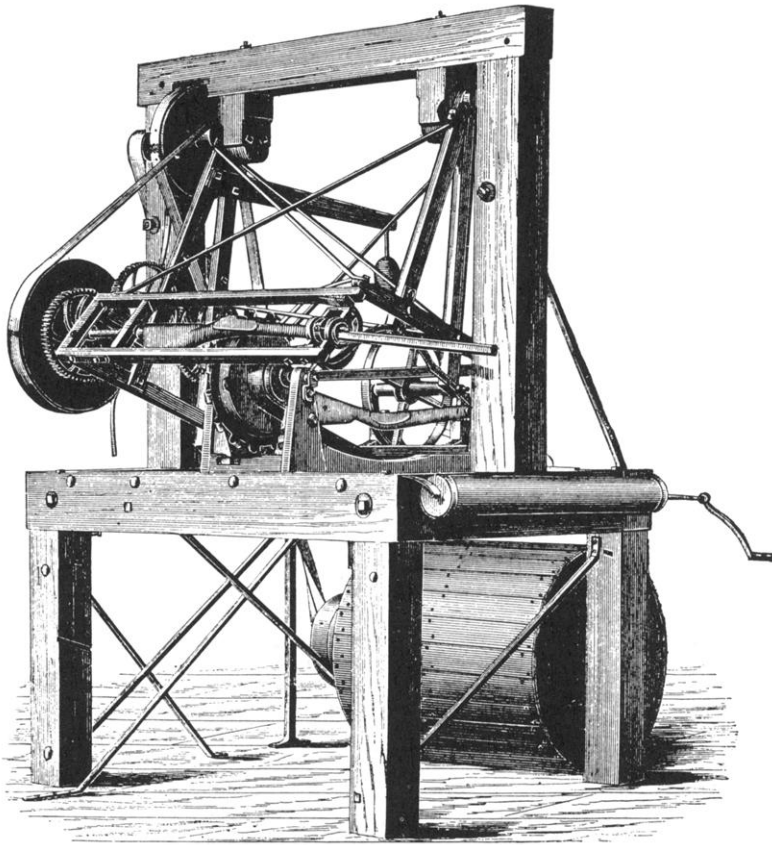


FIG. 4.—In the gunstock-turning lathe the gunstock and model hang parallel to each other in a double frame swinging on pivots overhead. The frame presses the gunstock model at rear against the tracer wheel and the gunstock blank against the rapidly rotating cutter wheel at front. The cutter and tracer wheels traverse simultaneously, cutting the contours of the slowly rotating stock from the butt to somewhat beyond the lock area, in about four minutes. (Charles H. Fitch, "Report on the Manufactures of Interchangeable Mechanism," *Tenth Census of the United States (1880)*, vol. 2, *Report on the Manufactures of the United States* [Washington, D.C., 1883], p. 629.)

for it was longer and more successful than most. But he demonstrated the full range of the behavior of an inventor as patent manager and can thus be regarded as a model for that role.

Interference and Reissue of Blanchard's Patent

Blanchard received a patent for his lathe in 1819 but only after winning an interference action brought by his neighbor in West Millbury, brass founder Asa Kenney, who claimed it was *his* invention.

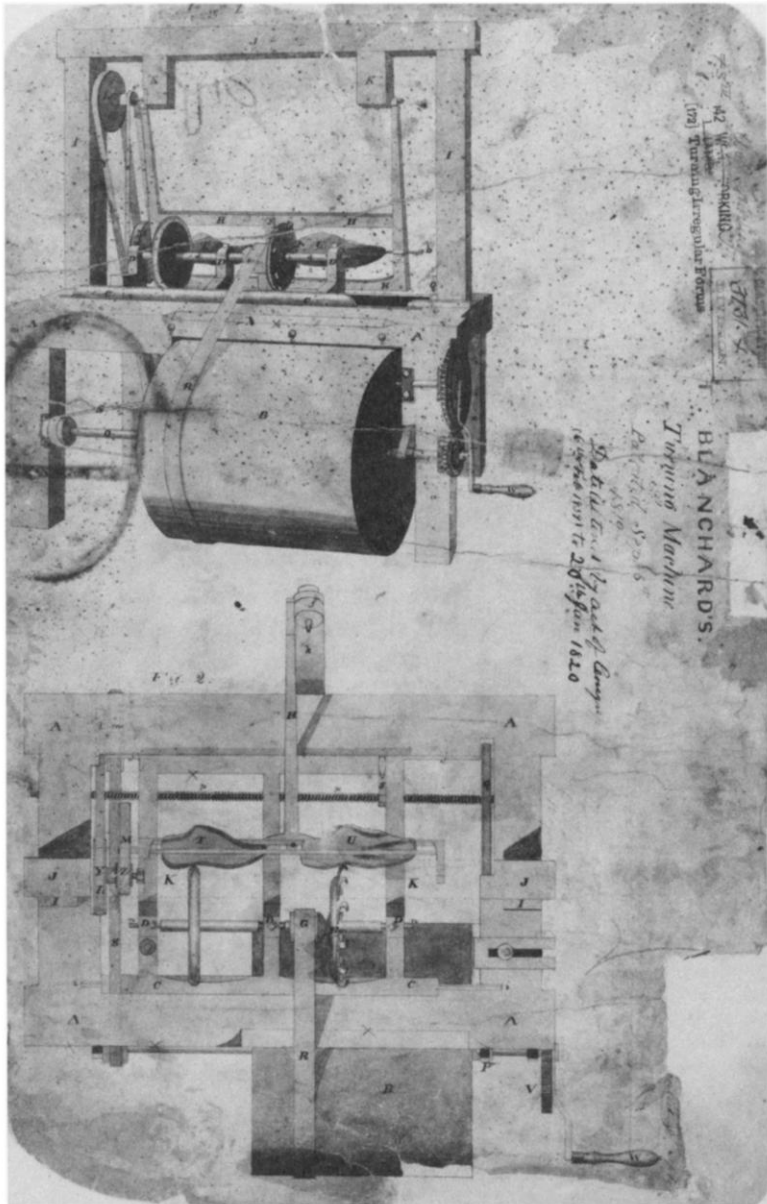


FIG. 5.—A drawing accompanied the written specification for a patent to help convey information about the invention first to the patent examiner and then to the public. Thomas Blanchard's original drawing for his irregular turning lathe presumably perished in the Patent Office fire of 1836, but this copy was restored to the record soon afterward. It shows an operating model of the machine copying a model last, *left*, to turn a rough workpiece, *right*, into a shoe last. (National Archives Cartographic Section, Alexandria, Va.)

An appointed panel of experts held a trial in Millbury in July 1819 and decided, despite the superior appearance of Kenney's brass model, that Blanchard's had precedence.²⁷ After obtaining his patent on September 6, 1819, Blanchard demonstrated a model of his machine in Boston and immediately ran into another patent-management problem. Some potential customers in Boston were scared off by other viewers, among them one William Hovey of Worcester, who questioned the machine's originality.²⁸ Blanchard withdrew his patent, rewrote his claims of originality in the specification, and obtained a reissue on January 20, 1820. The differences between the September and January versions of Blanchard's patent suggest that he informed himself in the interim about four other contemporary machines—for making pulley blocks, wool-card handles, ornamental turnings, and shoe lasts. His reissued patent specification ends by stating that "the principle of his machine or invention is different" from each of these other machines.²⁹

The last-making machine had been put into operation in Waterbury, Connecticut, by Azariah Woolworth, a sometime clockmaker, house carpenter, and ship carpenter.³⁰ In November 1819 Blanchard and his brother Stephen traveled to Waterbury to see it. Run partly by hand and partly by water, it was much cruder than Blanchard's machine and took about six times longer, but it was an improvement over Woolworth's earlier "machine #1," which was totally handpow-

²⁷John C. Crane, "Asa Holman Waters," in *Collections of the Worcester Society of Antiquity*, vol. 7 (1888), p. 85. Blanchard wrote to Springfield Armory Superintendent Roswell Lee on June 9, 1819, to tell him of Kenney's interference suit, to be decided by three commissioners, "and you are appointed [sic] by the Secretary of State as the principle [sic] one." Springfield Armory Records, Record Group 156, National Archives, Washington, D.C. On July 28, 1819, Millbury arms contractor Asa Waters debited Thomas Blanchard \$3.00 for "two days attending court." I infer this was Kenney's interference suit and that Waters probably testified as a witness. A. H. Waters Sr. and Son, Millbury, Ledger 1815–1820, p. 169, in Waters Family Papers at library of American Antiquarian Society, Worcester, Mass.

²⁸Henry Howe, *Eminent American Mechanics* (New York, 1840), p. 209. William Hovey, affidavit April 4, 1849, and Samuel Cox, affidavit March 11, 1849, both in Thomas Blanchard v. Isaac B. Eldridge, U.S. Circuit Court in Equity, Eastern District, Pennsylvania, October session 1848, Record Group 21, National Archives, Philadelphia Branch.

²⁹The 1819 and 1820 versions of Blanchard's patent specification were both restored after the December 1836 fire at the Patent Office and were copied into *Restored Patents*, vol. 4, 1817–22, pp. 297–302 and 349–63, respectively. The thirty-or-so volumes of *Restored Patents* are in Records of the Patent Office, Record Group 241, National Archives, Alexandria, Va.

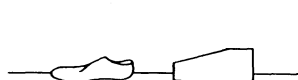
³⁰Azariah Woolworth, deposition of January 16, 1849, in Blanchard v. Eldridge (n. 28 above). Woolworth was a native of Longmeadow, near Springfield, Mass., according to Joseph Anderson, *The Town and City of Waterbury* (New Haven, 1896), 2: 454.

ered.³¹ Figure 6 shows four points of comparison between Woolworth's "machine #2" and Blanchard's irregular turning lathe. Each comparison shows both a similarity between the machines and a

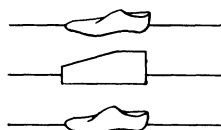
COMPARISON OF FEATURES

BLANCHARD'S IRREGULAR TURNING LATHE
(PATENTED SEPT. 6, 1819)

WOOLWORTH'S IRREGULAR TURNING LATHE "#2"
(OBSERVED IN AUTUMN 1819)

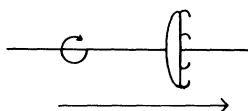


MODELS

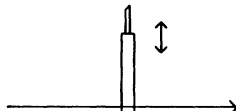


ONE MODEL, ONE WORKPIECE

TWO MODELS, ONE WORKPIECE



CUTTERS



POWERED ROTARY CUTTER
TRAVERSES BY POWER

POWERED RECIPROCATING CUTTER
TRAVERSES BY HAND



MOTIONS

MODEL AND WORKPIECE ROTATE
CONTINUOUSLY BY POWER

MODELS AND WORKPIECE ROTATE
INTERMITTENTLY BY HAND



EFFECTS



CUTS SPIRAL PATH AROUND WORKPIECE

CUTS LINEAR PATH ALONG WORKPIECE

FIG. 6.—A comparison of Thomas Blanchard's irregular turning lathe with the contemporaneous Waterbury last-making machine shows both similarities and differences in conception of how to copy a three-dimensional model by machine. (Drawing by Lyn Malone.)

³¹Woolworth's deposition, 1849.

difference. First, like Blanchard's machine, Woolworth's machine #2 used a tracing mechanism to copy a model, but instead of one model, it used two. Second, both machines differed from ordinary lathes in having separately powered cutters, but Blanchard's cutter was rotary and Woolworth's was reciprocating. Third, in both machines, the workpiece and model(s) were keyed together to rotate simultaneously, but in Blanchard's they rotated continuously, while in Woolworth's they rotated intermittently, at the end of each lengthwise cut along the workpiece. Finally, as a result of their respective actions, both machines were capable of turning out irregular objects like shoe lasts, but where Blanchard's lathe cut a single continuous spiral path around the workpiece, Woolworth's last-making machine cut a series of parallel paths lengthwise along the object. Overall, how would you weigh the differences against the similarities? Are the two machines "the same" or "different"?

Thomas and Stephen Blanchard considered them different. Stephen Blanchard told his brother he had "nothing to fear from *that* machine."³² Nevertheless Thomas took the precaution of revising his patent specification to deny any similarity between his lathe and the Waterbury last-making machine. Woolworth sold his unpatented invention to William Hovey and helped him build an improved version. It was much more like Blanchard's, having a powered continuous rotation of the workpiece and a single model, and an additional rotary sawtoothed cutter for smoothing off the edges left by the first cutter.³³ In early 1820 the financial backers of this Woolworth machine #3 were threatening to sue Blanchard and his backers, and vice versa. They "compromised" out of court: the Blanchard group bought all rights from the Woolworth group, and Woolworth's machine #3 received a patent in June 1820.³⁴ In 1834, when both patents expired, Blanchard got his extended, but Woolworth did nothing about his, so his machine #3 should have been in the public domain. A fire in the Patent Office destroyed all records in 1836, however, and no one restored Woolworth's patent specification to the record afterward. Blanchard had his own patent restored.

³²Stephen Blanchard, affidavit in *Thomas Blanchard v. Chandler Sprague*, U.S. Circuit Court Massachusetts District, 1838, Record Group 21, National Archives, Boston Branch.

³³The affidavit of Phineas Dow, August 10, 1848, in *Blanchard Gunstock Turning Factory v. Isaac Eldridge et al.*, U.S. Circuit Court Eastern District Pennsylvania, April Session 1848, includes a copy of Woolworth's patent specification. Record Group 21, National Archives, Philadelphia Branch.

³⁴On the compromise, see Samuel Cox's deposition of October 17, 1848, in *Blanchard Gunstock Turning Factory v. Isaac Eldridge et al.* (n. 33 above).

Blanchard's Assignments and Licenses

Meanwhile, once he obtained his reissued patent in January 1820, Blanchard was able to divide up his intellectual property in two ways: by its numerous uses—for making shoe lasts, tool handles, wheel spokes, and so on—and by geographical territories of various sizes. He sold and licensed his machine for the same use in different territories and for different uses in the same territory.³⁵

As a landlord decides among his choices of occupying, selling, and renting his house, so Blanchard had to decide about the long- and short-term advantages of using his own invention or assigning or licensing his patent. Eli Whitney, for instance, had made what proved to be the wrong decision in attempting to produce and operate cotton gins himself instead of licensing others to make and use them.³⁶ Ithiel Town, in contrast, was happy to let other bridge builders use his 1820 patented lattice-truss design and to collect from them royalties of \$1 per foot. (He sued them for \$2 per foot if they tried to cheat.)³⁷

On the other side of the bargains, Blanchard's assignees and licensees also made decisions. They had to weigh the short- and long-term costs of pirating, buying, or leasing the patent rights to make or use Blanchard lathes in a desired territory. They had to decide whether they wanted an assignment, with the right to issue licenses and to sue infringers on their own, or wanted instead to leave those tasks to Blanchard and take a license. And they had to decide from what size territory they wished to exclude competitors from operating. Other things being equal, the larger territories cost more.

The assignment records indicate that the result of these considerations by Blanchard and his assignees and licensees was a mixed strategy of his own use, plus assignments and licenses in territories ranging in size from the whole country to one "shop right" in one town. Blanchard, then, acquired several management roles—user,

³⁵The records for assignments of rights to Blanchard's various patents are scattered through the volumes of *Transfers of Patent Rights*, Libers A to Z in each of series 0 to 9, dating from ca. 1836 to 1870. These are held by the National Archives in the Federal Records Center, Suitland, Md. They are digested in *Digests of Patent Assignments*, "B" vol. 1 to 4, Record Group 241, National Archives. These digests, which also constitute the index to the complete assignments, are themselves indexed by the patentee's name in separate volumes, also in Record Group 241, National Archives, Washington, D.C. For procedures in using assignment records, see Cooper (n. 17 above).

³⁶Jeannette Mirsky and Allan Nevins, *The World of Eli Whitney* (New York, 1952), chaps. 8 and 9, discuss this choice and the tribulations of partners Miller and Whitney in attempting to follow it.

³⁷Joseph Gies and Frances Gies, *The Ingenious Yankees* (New York, 1976), p. 120. William P. Trowbridge, "Town's Truss," *Columbia University School of Mines Quarterly* 9 (July 1888).

seller, and licensor—in relation to his own intellectual property and to the other persons who used it. The network of assignees and licensees that grew up around the irregular turning patent formed a nascent interest group that could be mobilized on occasion to testify in lawsuits against infringers and help lobby in Congress. Over the decades the assignments of Blanchard's lathe for various purposes spread westward across the expanding nation (see fig. 7).

Extension of Blanchard's Patent

Blanchard managed to extend the duration of his patent twice by act of Congress, from its initial fourteen-year term to a total of forty-two years. Instead of normal Patent Office procedures for extension, this involved lobbying in Washington for passage of special legislation. To obtain the first extension, Blanchard argued that the fair value of his machine was at least \$215,440 but that his work at Springfield Armory had hindered him for several years in managing his patent, so he had netted only \$3,665.84 in the fourteen years' term.³⁸

Unfortunately for Blanchard, several months elapsed between the expiration of his patent in January 1834 and its renewal in June. The congressional act of renewal (6 Stat. 589) exempted from infringement suit (and from royalty payments) anyone who had not previously been using Blanchard's lathe but who had in good faith started doing so during those few months in which it was unknown that it would be renewed. This "free" period caused Blanchard trouble later: whenever an accused infringer claimed he had built his Blanchard lathe during those few months, Blanchard would have to provide evidence to try to disprove it. The renewal act also mistakenly dated Blanchard's 1820 patent as January 12 instead of January 20. In a later lawsuit, a defendant seized on this error, claiming that the renewal was not legally binding, since it referred to a nonexistent patent!³⁹ Blanchard had to go to the trouble of getting another act through Congress in 1839 (6 Stat. 748) to correct the act of 1834.

When the second term of Blanchard's patent expired in 1848, Blanchard managed better. He obtained renewal for an unprecedented third fourteen-year term by an act of Congress that left no dating gaps or defects (9 Stat. 683). To do so, he added showmanship to his patent-management behavior: he obtained plaster busts of influential congressmen and used a sculpture-copying machine to

³⁸Thomas Blanchard, "Petition to the Senate and House of Representatives," December 28, 1833, presented to the House January 6, 1834, pp. 4, 5. A copy made June 13, 1838, is in the file of *Blanchard v. Sprague*, 1838 (n. 32 above).

³⁹*Blanchard v. Sprague*, 1838; 3 Federal Cases: 645–46.

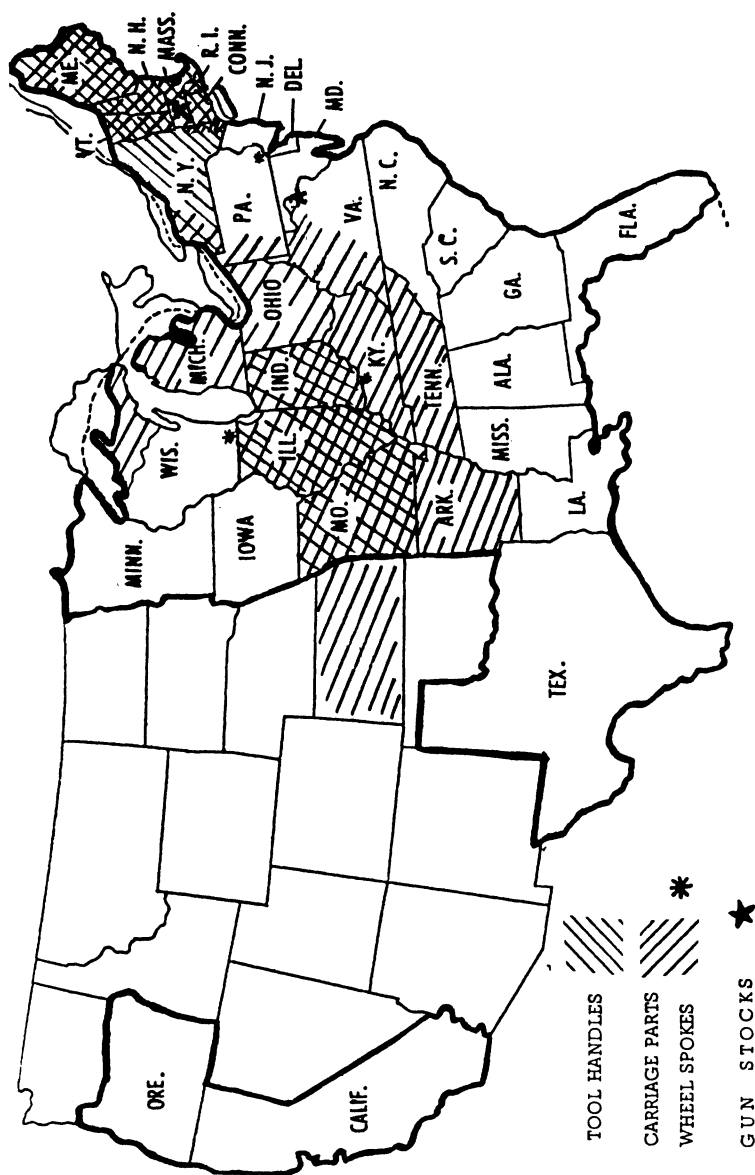


FIG. 7.—Blanchard's assignment for making or using his lathe for specific purposes in particular towns, counties, states, or combinations of states began in New England in the 1820s and spread southwest by 1860 to the territories shown. Stars represent the national armories at Harpers Ferry, Va., and Springfield, Mass.

make marble copies, which he exhibited in the Capitol. As the flattered congressmen admired these marble sculptures, Blanchard told them this was a new use of his irregular turning lathe, for which he needed a patent extension. This episode misleadingly added sculpture copying to the popular list of uses for the Blanchard lathe, even though the machine he used could not have been the Blanchard lathe whose patent they renewed, for its cutter would be unable to cut such fine detail as the features of a face.⁴⁰ The counsel for opposition to Blanchard's extension bill, Rufus Choate, who was a former representative and senator from Massachusetts, quipped afterward that Blanchard had " 'turned the heads' of Congress and gained his point."⁴¹

Not everyone was amused at Blanchard's 1848 extension. An outraged correspondent, James Johnston, wrote to the *Scientific American* to predict darkly that the extension would result in an intolerably "anti-republican" regime imposed on all manufacturers of shoe lasts and ax helves: they would have to let Blanchard's agents inspect their books and repossess machinery in case of default and would not "be allowed to dispose of any last by barter, nor receive any payment except money, nor give longer credit than six months." Johnston predicted that "the renewal of Blanchard's patent will thump so hard upon the heads of our Last Manufacturers that some of them no doubt will invent a machine shortly, that will successfully evade Blanchard's patent."⁴²

Soon afterward, another correspondent announced that Elbridge Webber of Gardiner, Maine, had indeed invented such a machine, which would appear in a few weeks, so that, "notwithstanding the broad ground which Mr. Blanchard has been allowed to enclose by a special act of Congress giving him a second renewal, it is hoped that he will not be allowed to make a turnpike of it and hold the office of tax-gatherer."⁴³ Webber's machine used as a model a "reverse pattern," whose contours a tracer followed, while another part of the machine reversed these into positive motions for cutting the workpiece. This

⁴⁰Presumably the sculpture-copying machine Blanchard used was the one for which he later obtained French Patent no. 14,290 in August 1855. It had a small vertical-spindle rotary cutter instead of the large horizontal-spindle cutter of his lathe. *Descriptions des machines et procédés pour lesquels des brevets d'invention ont été pris sous le régime de la loi du 5 juillet, 1844*, 2d ser., vol. 51 (Paris, 1855), p. 186.

⁴¹[Asa H. Waters], "Thomas Blanchard the Inventor," *Harper's New Monthly Magazine* 68 (June–November 1881): 258.

⁴²*Scientific American* 3, no. 31 (April 22, 1848): 245.

⁴³*Scientific American* 3, no. 34 (May 12, 1848): 270.

blatant attempt at evasion was rejected by the patent examiner as a claim to “invention.” The machine received a patent in 1849, however, on the basis of two other claims.⁴⁴

Litigation over Blanchard's Patent

Like other petitioners for patent extension, Blanchard said he had been unable to earn a reasonable return on his irregular turning lathe during its original patent term because people had used it without paying him and he had had to spend so much on lawyers' fees defending his patent in court. Reportedly “more than fifty violators had pirated his invention” during his years at Springfield Armory.⁴⁵ From the 1820s through the 1850s, Blanchard and his assignees sued dozens and dozens of users of copying lathes for making spokes, lasts, and tool handles.

Despite Webber and other new “evaders” of his patent, Blanchard not only continued to win most of his court cases against them, he also won the right to collect royalties after the extension in 1848 from those who had first built their machines in those “free” months in 1834 and had therefore never paid royalties before.⁴⁶ He failed, however, in the attempt to collect additional royalties from Eli Whitney, Jr., who had bought a stocking lathe in 1843 from Blanchard himself, for a price that included royalties for the then-remaining term to 1848. In the case of *Thomas Blanchard v. Eli Whitney*, Justice Samuel Nelson reasoned that in “a case where the patentee has manufactured and sold the machine himself . . . such a transaction . . . implies and carries with the purchase of the article the right to use it, at least till it is worn out.”⁴⁷

Frequently the “infringing” users were licensed by patentees of other irregular turning lathes that were improved versions of Azariah Woolworth's Waterbury last-making machine. For despite his brother Stephen's reassurance in 1819 that Woolworth's machine was nothing to worry about, its descendants kept cropping up, especially in Connecticut, with various modifications that made them more like Blanchard's lathe and more of a competitive threat to Blanchard's

⁴⁴See correspondence in patent application files for U.S. Patent no. 6,253 granted April 3, 1849, to Elbridge Webber and Charles Hartshorn, Record Group 241, National Archives, Suitland, Md.

⁴⁵D. Hamilton Hurd, ed., *History of Worcester County, Massachusetts* (Philadelphia, 1889), 2:976.

⁴⁶*Thomas Blanchard v. John Haynes*, U.S. Circuit Court New Hampshire District, 1848; 3 Federal Cases: 628–29.

⁴⁷*Thomas Blanchard v. Eli Whitney*, U.S. Circuit Court Connecticut District, September 1855; 3 Federal Cases: 651.

machine in performance. Some of these received patents, in spite of Blanchard's vociferous protests to the Patent Office that they were merely "useless or *nonimproved* modes of endeavoring to evade his patent."⁴⁸ Figure 8 shows one descendant that appeared in France, beyond Blanchard's range of litigation.

Witnesses in Blanchard's lawsuits were called on to testify whether the two machines—Blanchard's and the defendant's—were substantially "the same" or "different" from each other. The witnesses identified themselves as mechanics, machinists, engineers, millwrights, inventors, last makers, ax-handle makers, wheel-spoke makers, shoemakers, gun makers, carriage makers, patent agents, and former patent examiners. Different features of the machines struck different witnesses as decisive. Some thought it was enough to say that both made the same product, so they were the same machine. Some said both machines copied models, so they were the same; others said one machine required two models and the other required one model, so they were different. Some said reciprocating cutters were of a different form from circular cutters, so they were different machines. Others said differences in the cutter form were immaterial; the machines were mechanically equivalent. Some said the cutters followed different paths on the workpiece, so they were different. Some said the one machine could or could not do smoother work than the other, or make more copies at once, or make larger or smaller sizes than the models, so the machines were substantially different or the same.

Over time, some arguments emerged as more persuasive than others. Evidence for sameness that sounds naive to us today, such as the fact that both machines made lasts, gradually dropped out. Distinctions that now sound more sophisticated stayed in use. For example, the more-or-less expert witnesses tended over time to emphasize process more than product and the kinematics of the machines more than their components. This gradual process, which was taking place over the very period in which there was a great proliferation of mechanical experience and increase in mechanical skills by people making and using new machines for many purposes in this country, is an example of social construction, at a deeper level, of the decision rules for invention itself in the realm of machinery.

The increasingly mechanically informed community included patent agents and patent examiners, as well as makers and users of

⁴⁸See, e.g., Timothy Clark, U.S. Patent no. 4,932, January 19, 1847; Smith Beers, U.S. Patent no. 7,806, December 3, 1850; Philo Beers, U.S. Patent no. 7,937, February 18, 1851. Blanchard's protest (with original emphasis) is in the application file for Philo Beers's patent, Record Group 241, National Archives, Suitland, Md.

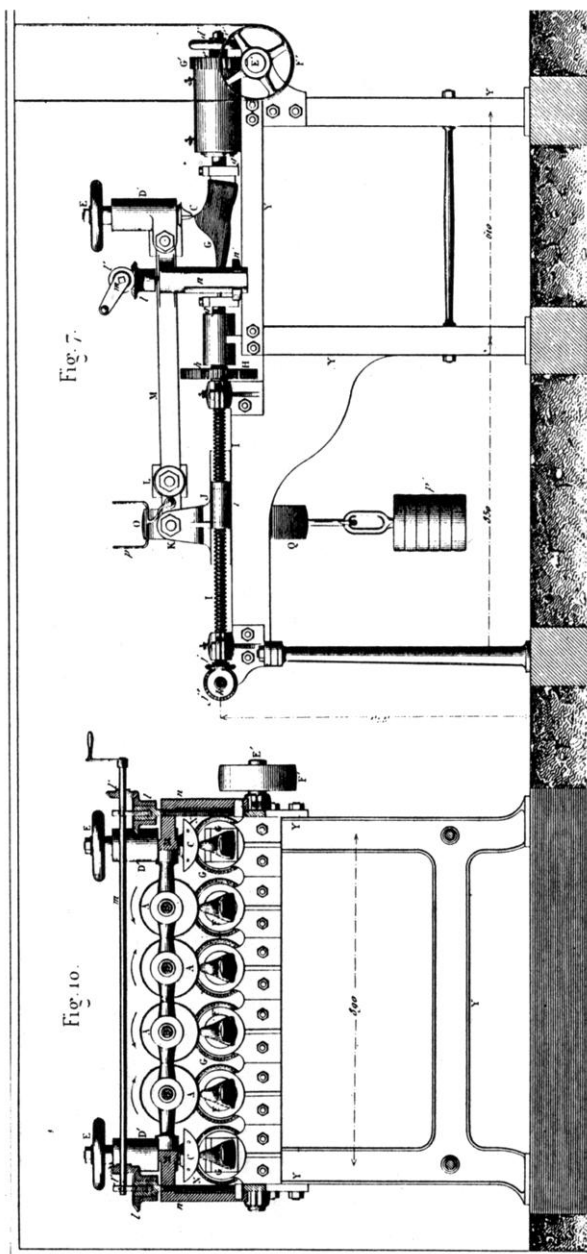


FIG. 8.—This French machine of the 1850s is a descendant of Azariah Woolworth's last-making machines #2 and #3 (of 1819–20) in that it uses two model lasts (G, far right and far left in the end elevation) and two feelers (C) to control the rising and falling of the rotary cutting wheels (A) over the four workpieces in the middle. (Armengaud Aine, *Publication industrielle des machines, outils et appareils* . . . , vol. 7 [Paris, 1856], pl. 9.)

machines. They all put their changing rules for judging originality into practice when called on in lawsuits or when communicating in other contexts, such as patent applications. The implicit rules for originality also gradually became more refined in subcategories where there were more machines to be compared with one another. Patent Office reclassification was one manifestation of this social construction. Where the Patent Office recognized twenty-two classes for patents for most of the 19th century, in 1870 it recognized 145 classes, because of subdivision of former classes as well as creation of new ones. (Today the Patent Office employs a whole division for continual reclassification.)

At a more particular level, over time a body of belief or understanding or knowledge concerning the specific machines at issue was also built up within the social subsystem for patent management, defining what Blanchard's particular "invention" had been and therefore what his intellectual property currently was, for which he deserved to enjoy a temporary monopoly. Some features were added to Blanchard's lathes in actual operation over the years, such as the arrangement of levers for making several sizes of well-proportioned lasts or hat blocks without changing the model (see fig. 9). One of Blanchard's licensees developed this feature around 1835 but never bothered to claim it as *his* intellectual property, and it came to be regarded as an integral part of the Blanchard lathe. Its integration into the lathe was confirmed during lawsuits, despite a difference of opinion at first as to whether the wording of Blanchard's specifications actually covered that device.⁴⁹ This gradual, repeated process is an example of the social construction of a particular invention. In this process of social construction, the boundaries of Blanchard's intellectual property—his lathe—expanded to subsume many similar machines, including, strangely enough, a machine that Blanchard originally said his lathe 'did *not* resemble: the Waterbury last-making machine.

Isaac Eldridge's Machines

A dramatic demonstration of this expansion took place when Blanchard sued several last makers in Philadelphia in the late 1840s. The court enjoined one of the defendants, Isaac Eldridge, from using his lathe. But during the trial Eldridge heard testimony by the elderly

⁴⁹Collins Stevens, depositions in *Blanchard v. Sprague*, 1838 (n. 32 above), and on September 16, 1856, in *Thomas Blanchard v. Warren Wadleigh and Isaac Lane*, U.S. Circuit Court New Hampshire District, 1857, Record Group 21, National Archives, Boston Branch.

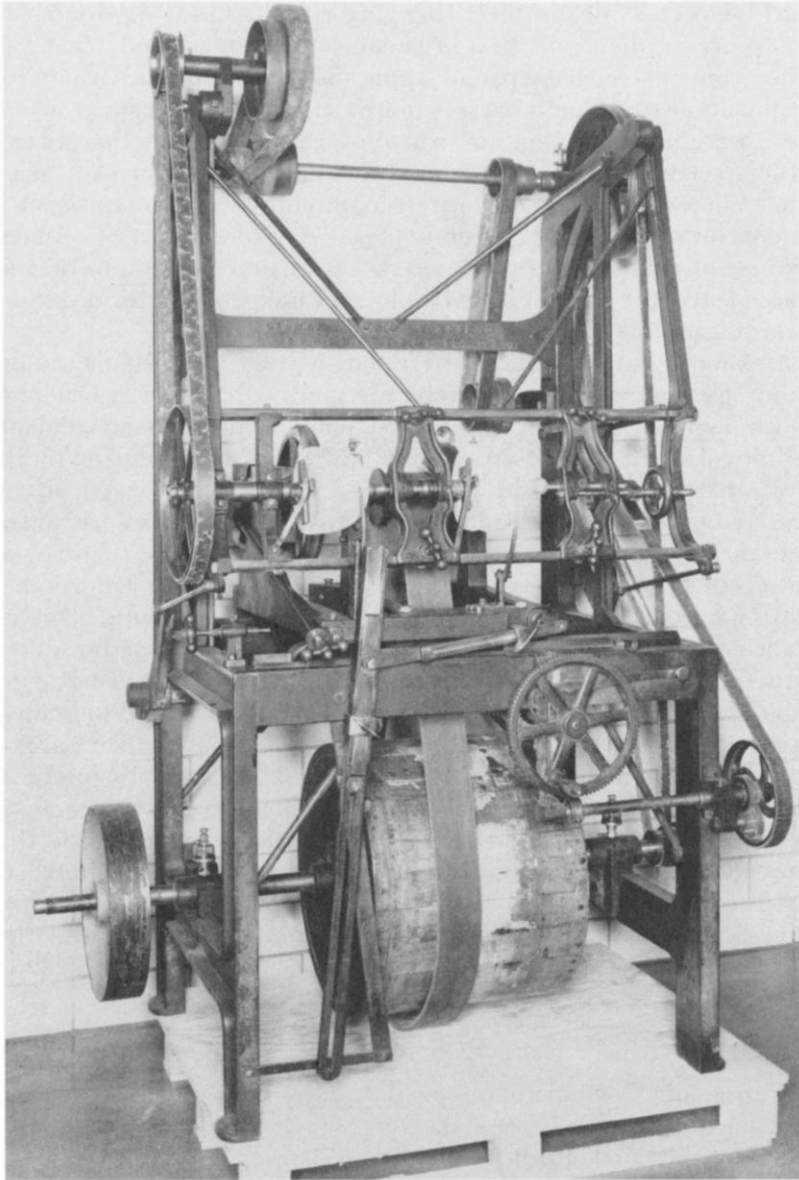


FIG. 9.—This hat-block lathe of ca. 1870, manufactured by Gilman and Townsend in Springfield, Vt., could cut proportionally correct hat blocks of several different sizes on the right, following the same model on the left. The levers and gears for accomplishing this were built into Blanchard lathes by assignees and licensees in the 1830s and gradually became accepted as a feature covered by Blanchard's patent. (National Museum of American History.)

Woolworth himself and others who described Woolworth's machines #1, #2, and #3 of 1818–20. Eldridge “experimented upon the vibrating cutter, as in a certain machine called at the trial of this case No. 2,” and built himself a version of the Waterbury last-making machine, with a reciprocating rather than rotary cutter.⁵⁰ With it he resumed making lasts, reasoning that although it was slower and less efficient it was surely safe from an accusation of infringement, since Blanchard's specification *said* it was different.

The court commissioned Philadelphia engineer W. W. Hubbell to report on the features of Eldridge's new machine. During Eldridge's demonstration for Hubbell, the building shook, the power transmission belt broke three times, and a bearing caught fire, but Hubbell saw enough to decide that the machine differed from a Blanchard lathe only in having a double-edged reciprocating cutter instead of a rotary one and a “friction column” as a tracer instead of a friction wheel.⁵¹ This was insufficient difference to satisfy Judge John K. Kane, although he did compliment Eldridge for his “considerable mechanical ingenuity.” Remarking that “there appears to be but a single point of difference”—the cutter—and that the two types of cutter were in principle the same, he issued a second injunction to make Eldridge stop using his new machine, too. Judge Kane said, “The patent law would give but an illusory protection to the meritorious inventor, if it respected devices like this.” He scolded Eldridge for spending his “inverted energies, not to improve or advance, but to devise something less useful and more costly than that which was known before.”⁵² Clearly, the judge was not about to tolerate this inelegant machine at its face value but was taking Eldridge's evasive motivation into account.

Perhaps rashly, Judge Kane went on to express “the essential principle that resides within” Blanchard's invention: “A tracer so arranged as to pass in a spiral or helix line over the surface of a model while the rough material revolves in a similar line under a cutter, guided by the tracer, but acting with independent, rapid motion—the combination of these for a declared purpose, this is the principle of the Blanchard patent. All the rest is detail.”⁵³ Picking up on this explanation, the invertedly energetic Eldridge rebuilt his machine so that the circular and lateral motions causing the spiral path of the cutter on the workpiece were no longer simultaneous, but sequential,

⁵⁰Isaac Eldridge affidavit, February 21, 1849, in *Blanchard v. Eldridge* (n. 28 above).

⁵¹W. W. Hubbell, master's report, February 27, 1849, in *Blanchard v. Eldridge*.

⁵²Judge Kane's opinion appears in 3 Federal Cases: 622–23, and in *Scientific American* 4, no. 35 (May 19, 1849): 27.

⁵³*Ibid.*

and resumed operation. Now his machine cut lengthwise paths on the lasts. Encountering Blanchard and his assignee A. K. Carter in Boston, Eldridge taunted them, saying "they could not stop me," for "Judge Kane had very properly decided what was Mr. Blanchard's principle and that was more than any other judge had done, which was the cutting in a spiral line."⁵⁴

The court next sent a blue-ribbon commission of three eminent members of the Franklin Institute to inspect Eldridge's third machine. They reported that it had a cutter wheel and a tracer that passed "rapidly from one end of the model to the other and backward . . . and at each end of the motion the model and rough material receive a small and equal motion around their longer axes." So Eldridge's machine now resembled Woolworth's machine #2 not in the cutter but in the intermittent rotation of the workpiece. The commissioners pointed out that these were different kinematics from those set forth in Blanchard's specification: "The only method proposed by Mr. Blanchard is that in which the friction wheel or tracer describes a spiral over the whole surface of the model and causes the cutters to act in a similar direction." They unanimously concluded that Eldridge's method "is not a mere colourable and unimportant change from the method described in Mr. Blanchard's patent, but that it is essentially different" and that therefore "the machine of Mr. Eldridge is different in its principle and mode of operation from that described in Mr. Blanchard's patent."⁵⁵ On the basis of this report, the court refused Blanchard's motion for attachment of Eldridge's property.⁵⁶

Taking Eldridge to trial yet again, to seek an injunction against his third machine, Blanchard not only marshaled additional expert witnesses to contradict the commissioners' report and testify to the mechanical equivalence of the continuous and intermittent motions. He also brought into court a model last-making lathe that was capable of either motion, by the shifting in or out of a simple mechanical linkage. It could perform in the Blanchard-specified continuous, spiral-forming fashion *or* in the Woolworth mode, with intermittent rotation of the workpiece between lateral lengthwise passes of the cutter.

⁵⁴Isaac Eldridge affidavit, October 5, 1849, *Blanchard v. Eldridge* (n. 28 above).

⁵⁵John F. Frazer, John C. Cresson, and Charles B. Trego, "Report of the Commissioners," December 1, 1849, in *Blanchard v. Eldridge*. For credentials of the commissioners, see Bruce Sinclair, *Philadelphia's Philosopher Mechanics: A History of the Franklin Institute 1824–1865* (Baltimore, 1974), pp. 254–58 and *passim*.

⁵⁶*Equity Docket Books* 3 (1848–58), p. 25, U.S. Circuit Court Eastern District, Pennsylvania, October Session, 1848, Record Group 21, National Archives, Philadelphia Branch.

Justice Robert Grier, the judge in this case, complained of the hot weather and said, "I should have had much less difficulty in arriving at a conclusion satisfactory to my own mind in the present case but for the opposite opinions expressed by gentlemen of the highest reputation for learning, judgment, and practical skill in mechanics." Although tempted to leave the question to a jury in yet another trial, the judge granted the injunction. He based his opinion on an analysis of the spiral motion into the separate lateral and circular motions of the cutter and workpiece, respectively, as Blanchard's courtroom model had demonstrated could be done. He concluded that "such a change . . . affecting the motions of the model and guide only in the figure of their path or the relative lines of their movements in no case changes the principle . . . of the machine." He added, "We cannot shut our eyes to the fact that the defendants have pirated the invention . . . in all its essential parts."⁵⁷ Like Judge Kane, Justice Grier was also taking Eldridge's evasive motivation into account, in seeing one Woolworth-like feature after another as "the same" as those in a Blanchard lathe. The irregular turning lathe had become "the Blanchard lathe," certainly *not* "the Woolworth lathe."

Effects of Patent Management on History

Thus, under treatment as a piece of intellectual property, Blanchard's invention had expanded in concept to include by 1850 all the features of a machine that Blanchard had explicitly disclaimed at the beginning in 1820. Presumably Blanchard had also changed his perception of the similarities and differences between Woolworth's machine and his. In those thirty years, the accepted notion of Blanchard's rights as intellectual property owner had reshaped the social perception of what his invention had been. I consider it likely that such a social redefinition was taking place, perhaps less dramatically, in many other cases also and that our history of technology has been distorted when two initially separate inventions can be regarded as one invention a few decades later.

This episode demonstrates especially clearly that the definition of any given invention and its delineation as intellectual property were the result of a social process that was in continuous operation, not just during the drawing up and approval of patent specifications, but also throughout the duration of the patent. It also suggests this process had a built-in bias against cheap (royalty-free) operation of obsolete,

⁵⁷Justice Grier's opinion in *Blanchard v. Biddle Reeves, Isaac Eldridge, et al.*, U.S. Circuit Court in Equity, Eastern District, Pennsylvania, September Term 1850, is reported in 3 Federal Cases: 639, 640.

less efficient machinery like Woolworth's machine #2. This bias may be seen as reinforcing market forces against technological inefficiency and against the small operators like Isaac Eldridge who might otherwise find economically viable niches using old-fashioned equipment. Thus, from the point of view of the society as a whole, vigorous patent management probably had a marginal "progressive" or ratchet effect on the development of technology. From the point of view of history, this bias had the effect of suppressing later recognition of how diverse the devices had been that were used in the past to achieve given ends, for example, last making. It therefore reinforced a misleading "great man" theory of invention.

Blanchard's Wood-bending Machine

In the case of Blanchard's wood-bending machine, for which he obtained U.S. Patent no. 6,951 in 1849, the outcome of social construction through patent management was quite different; one invention did not always subsume another to become the eponymous winner in the contest. Blanchard made more money but less fame from his bending machine than he did from his irregular turning lathe. The bending machine was in a very general sense like the irregular turning lathe, for it shaped a workpiece to a pattern, but it did so by bending it instead of cutting it. As shown in figure 10, it wrapped a steamed stick of wood around a rotating form or pattern. With a screw the operator compressed the stick lengthwise before bending and then eased off, so that as they were bending, the wood fibers never had to stretch beyond their natural limit. Thus Blanchard's machine prevented breakage of the wood, which had been happening two out of five times with previous methods. It was originally intended for bending plow handles, but Blanchard and his assignees also applied it to the bending of wheel felloes (rims) and other carriage and wagon woodwork, parts of furniture, and even ship timbers up to 14 inches thick.⁵⁸ With two semicircular bent felloes instead of six or seven shorter sawed ones, a wheel could be made lighter with no loss of strength. By bending ribs and knees from straight timbers, builders of clipper ships no longer needed to search for naturally crooked timbers.

Although the ship-timber enterprise failed commercially within a few years, the use of Blanchard's bending machines spread westward with agricultural implement manufacture and with the carriage and wagon industry, even as did Blanchard lathes for turning tool handles,

⁵⁸Blanchard's additional bending patents were no. 15,944 in 1856; and no. 19,480 and no. 20,137 in 1858.

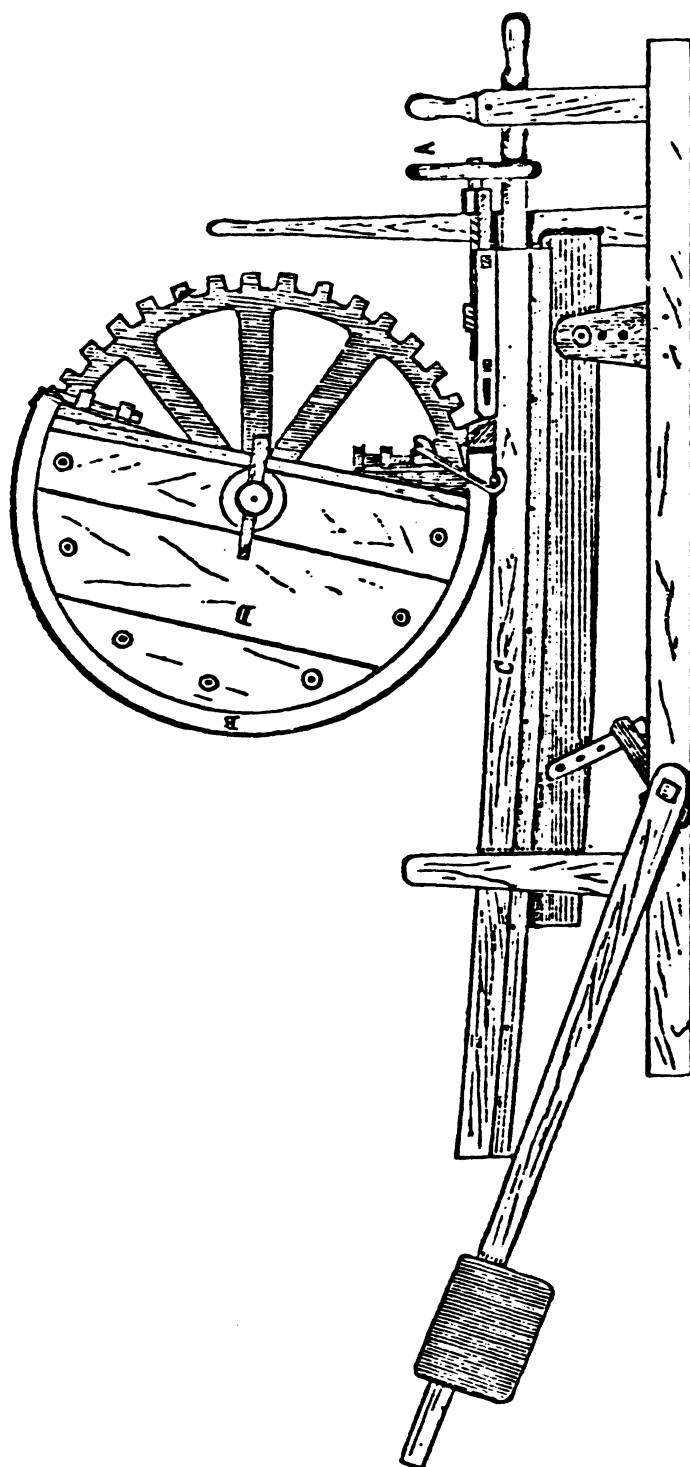


FIG. 10.—To bend a wheel felloe, Thomas Blanchard's 1849 patented machine first compressed a steamed stick, or workpiece *B*, lengthwise by means of the screw *A*, then rotated the form *D* to which the workpiece was clamped. With their ends trimmed off to 180 degrees each, two of these made up the rim of a wheel for a carriage or wagon. (*The Hub* 20 [1878]: 388.)

wheel spokes, and other irregular parts of carriages (see fig. 11). Some of his assignees used both the Blanchard lathe and his bending machine to make wheels or other parts of carriages and wagons. In managing Blanchard's patent, his assignees had, of course, to sue infringers.

Morris's Wood-bending Machine

In 1856, John C. Morris of Cincinnati, which was by then a center for manufacturing carriage parts, obtained U.S. Patent no. 14,405 for an "Improved Method of bending wood." In Morris's machine the form remained stationary and the steamed stick of wood was bent to it. By hand methods, such bending predated both Morris's and Blanchard's machines. Morris's machine did not compress the wood endwise in advance, but it did confine it endwise, so during bending some compression took place on its inner edge (see fig. 12).

Morris's assignees and Blanchard's assignees began a series of suits and countersuits. Presumably in response to the lawsuits, in 1859 Blanchard withdrew his patent, reworded the specification with slightly broader language, and obtained a reissue. Morris countered by obtaining a reissue of *his* patent in 1862, in which he distinguished two types of bending: "inward" and "outward," as shown in figure 13. He very much expanded his claim to originality within the class of "outward"-bending machines, which he had just defined. How would you judge the similarities and differences of the two bending machines? Are they the same overall or different?

At the end of 1862—that is, the year in which his 1820 irregular turning patent finally expired—Blanchard unsuccessfully petitioned the House and Senate for a fourteen-year extension of his 1849 wood-bending patent.⁵⁹ A year later, he obtained a seven-year extension by normal Patent Office procedure, having argued that his bending invention had very much benefited the shipbuilding industry, but had netted him little. His petition alludes to "my infirmities of old age" and to "a sense of duty to . . . my heirs."⁶⁰ It was possibly his last act of patent management, for he died four months later, in April 1864. As a piece of property, the wood-bending patent survived him until December 1870, in the hands of his assignees and *their* assignees.

After losing to the Morris faction at the Circuit Court level in 1867, the Blanchard faction finally won the wood-bending suits at the Supreme Court level. The upshot of the whole proceeding was to blur

⁵⁹Papers of Senate patent committee, 35th Cong., Record Group 46, National Archives, Washington, D.C.

⁶⁰Thomas Blanchard, extension application for wood-bending patent, December 17, 1863, Record Group 241, National Archives, Washington, D.C.

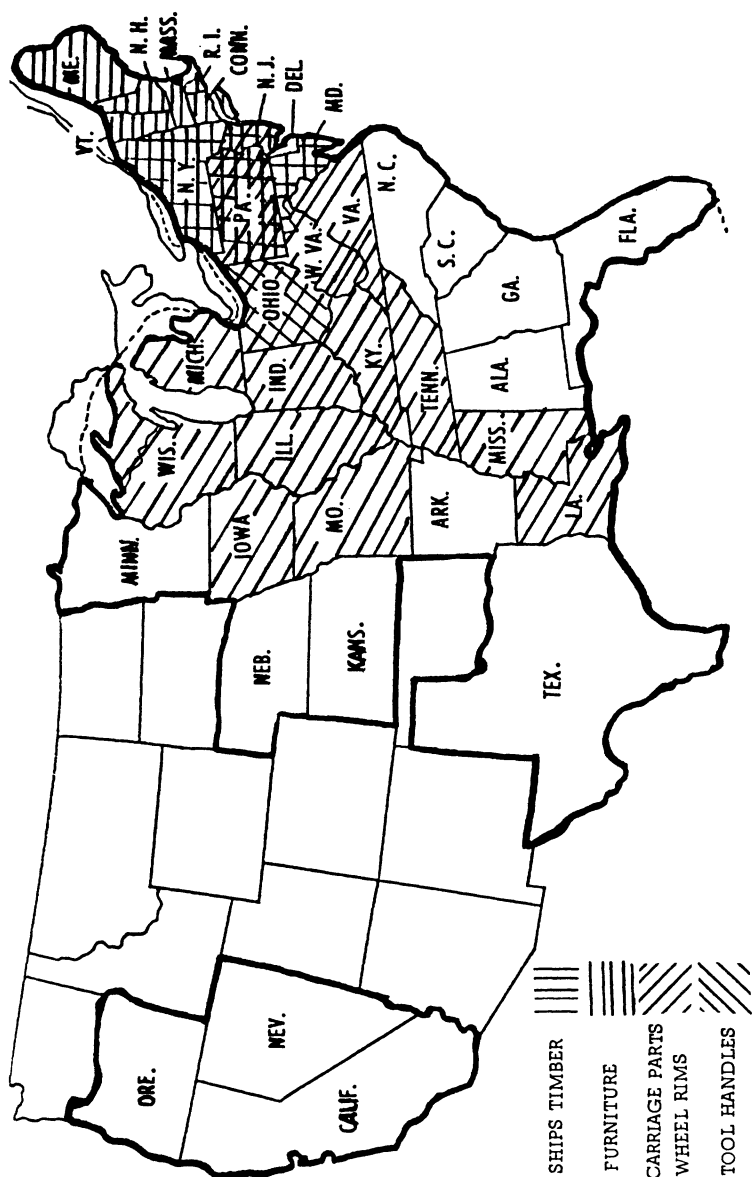


FIG. 11.—During its patent period of 1849 to 1870, the assignment of the right to make, use, or sell Blanchard's bending machine for various purposes spread over the territories shown.

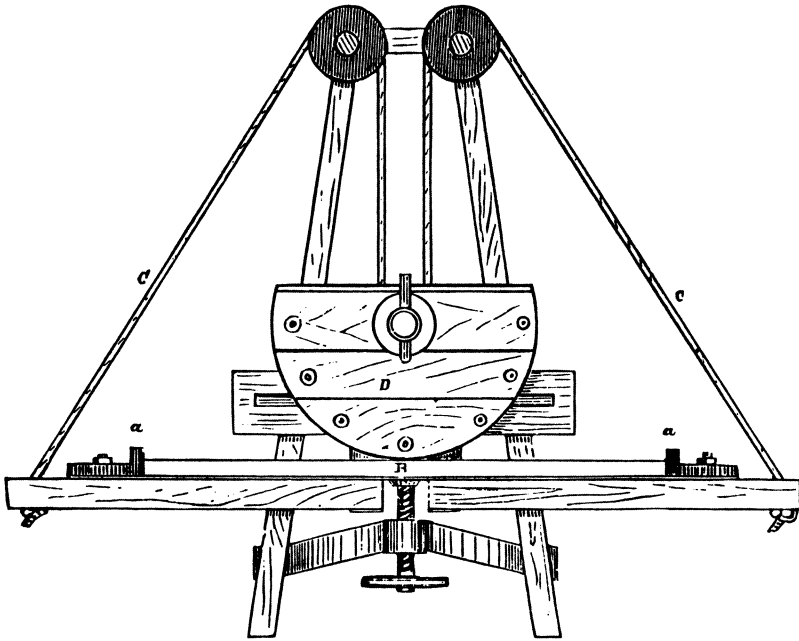


FIG. 12.—John C. Morris's machine for bending wheel felloes, patented in 1856, constrained the workpiece *B* within stops *a* but applied no end pressure beyond what resulted as it wrapped both ends of the workpiece up from the middle to curve around the stationary form. (*The Hub* 20 [1878]: 388.)

the identification of mechanized wood-bending with either Blanchard or Morris. Both types of machine were built beyond 1870 and were still used in wood-wheel works surviving until recently.⁶¹ This contrasts with the case of the irregular turning lathe, in which the social definition of Blanchard's invention eventually subsumed Woolworth's. In both cases, however, the implicit rules for deciding between two machines for achieving the same end—a shaped shoe last or a bent wheel rim—were socially constructed within the system of patent management. Such constructions were the outcome of human perceptions of differences and similarities, as affected by their views of property rights.

Thomas Blanchard was by all accounts a successful inventor. His other patented inventions ranged widely, from steam-driven carriages and riverboats to blockmaking machinery, felt hats, grinding mills,

⁶¹The machinery and operations of Hoopes, Brother, and Darlington of West Chester, Pennsylvania, were documented by a Smithsonian team in 1969. See Peter Haddon Smith, "Industrial Archeology of the Wood Wheel Industry in the United States" (Ph.D. diss., George Washington University, 1971).

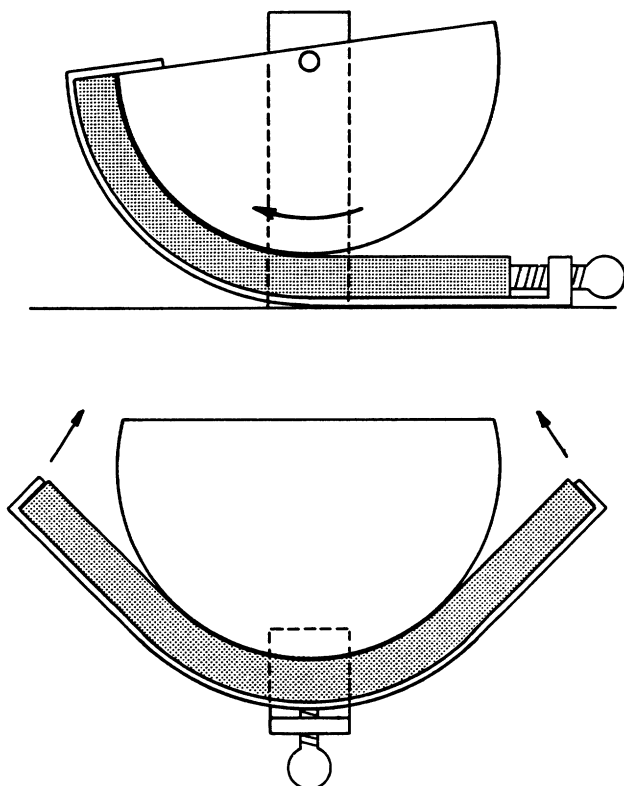


FIG. 13.—The Blanchard method of bending wood is shown above; below is the Morris method. In his reissued patent specification, Morris called his method “bending outward” and Blanchard’s “bending inward.” (Drawing by R. B. Gordon.)

scoop shovels, and even cigarettes. Through his assiduous practice of patent management, Blanchard grew affluent enough to move his family to a fashionable house opposite the Boston Commons and to take a trip to Paris in 1855, where he displayed his wood-bending and sculpture-copying machines at the Paris Exposition and won prizes for them. At the age of seventy-four he was included in the pantheon of American scientists and inventors portrayed by Christian Schussele in 1862, shown in figure 14.

Should we conclude that Blanchard deserved fame and fortune for inventing exceedingly useful and versatile machines? Or that he simply managed his intellectual property more adroitly than less “successful” inventors like Woolworth? Regardless of how we judge Blanchard’s personal achievement, we should recognize that his story reveals patent management as the social system in which people



FIG. 14.—American “Men of Progress,” 1862. In his seventies, Thomas Blanchard appeared, second from right, among these scientists, inventors, and entrepreneurs fictitiously assembled beneath Benjamin Franklin’s presiding image. Blanchard is turned toward Elias Howe. The others are, left to right, William Morton, James Bogardus, Samuel Colt, Cyrus McCormick, Joseph Saxton, Charles Goodyear, Peter Cooper, Jordan Mott, Joseph Henry, Eliphalet Nott, John Ericsson, Frederick Sickels, Samuel Morse, Henry Burden, Richard Hoe, Erastus Bigelow, and Isaiah Jennings. (John Sartain engraving after painting by Christian Schussele, 1862, National Portrait Gallery, Smithsonian Institution, Washington, D.C.)

continuously constructed and intertwined definitions of particular inventions with considerations of property.

Major studies have described the patent-management behavior, in an era later than Blanchard's, of inventors George Eastman, Thomas Edison, and Elmer Sperry.⁶² Other case histories have pointed to the social construction or shaping of particular technologies, say, bicycles, refrigerators, or turbojet engines.⁶³ Some of these studies describe overlaps and hierarchies of social subsystems in which social construction took place.⁶⁴ The patent-management system in which Thomas Blanchard and his contemporaries operated was also one—a very central one—of these overlapping subsystems that mediated between the individual inventor and the whole society. Overlapping with and drawing on the society's legal system, its economic system, and its political system, it had a more-or-less direct influence on what in particular an inventor chose to invent. It also continued to shape the invention throughout its patent period.

In the deeper, definitional sense of "invention," however, the patent-management system had an additional special relationship to American society's store of technological knowledge, for it was the arena in which the implicit rules were determined for deciding what was new. As Thomas Blanchard's experience illustrates, the interactions of persons during patent management brought about social decisions that identified not only who to reward for particular inventions but also what to remember as important differences or dismiss as unimportant samenesses between items of technology. Patent management was the subsystem to which our society had allocated this task of constructing "invention" itself. The subsystem not only guided inventors' technical choices; it shaped our very ways of perceiving technological objects.

Traditional Eskimos are said to distinguish many more varieties of snow than can modern Americans not because their innate sense perceptions are more acute but because it is more important to them

⁶²Reese Jenkins, *Images and Enterprise: Technology and the American Photographic Industry 1839 to 1925* (Baltimore, 1975); Robert Friedel and Paul Israel, with Bernard S. Finn, *Edison's Electric Light* (New Brunswick, N.J., 1986); Thomas Parke Hughes, *Elmer Sperry, Inventor and Engineer* (Baltimore, 1971).

⁶³Wiebe Bijker, Thomas Hughes, and Trevor Pinch, eds., *The Social Construction of Technological Systems* (Cambridge, Mass., 1987); Donald MacKenzie and Judy Wajcman, eds., *The Social Shaping of Technology* (Milton Keynes, 1985).

⁶⁴See, e.g., Ruth Schwartz Cowan, "The Consumption Junction: A Proposal for Research Strategies in the Sociology of Technology" (pp. 261–80); and Edward W. Constant II, "The Social Locus of Technological Practice: Community, System, or Organization" (pp. 223–42), both in Bijker et al. (n. 63 above).

to make those distinctions, so their culture has taught them to perceive those differences. Our cultural distinctions among man-made objects also reflect what is important to us—such as private-property rights. We will better understand our relation to technology and its history when we better understand how people have made technological distinctions within the patent-management subsystem of our society.